Technische Universität Hamburg

Data Science Foundations

Low-dimensional Control under Embodiment Constraints: From LQR to Reinforcement Learning

Bachelor's Thesis
am Institut Data Science Foundations
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vorgelegt von
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am
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Abstract

Your English abstract here (mandatory if written in English and recommended otherwise).

Zusammenfassung

Hier die deutsche Zusammenfassung einfügen (notwendig).

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Chapter 1

Introduction

The introduction describes the problem, why it is important, the main ideas of the thesis, what are the main contributions of the thesis, etc. The introduction is the most important part of your thesis because it answers the question of why you are researching your topic at all. The introduction does not only motivate your thesis, it also justifies it. Compared to the rest of your thesis, it is a very short text, but this text should be extremely well-written and clear. It deserves a lot of polishing. If the introduction is boring or not well written, a reader (and your supervisor who will finally grade your thesis) will most likely not have fun reading your work. The introduction (together with abstract and title) is the "bait" to get your reader, your supervisor and your reviewer hooked.

The introduction should be a short version of the whole thesis, with a focus on motivation. A good introduction is structured as follows:

1. It should start with a brief motivational problem outline to set the stage, e.g., "It is important to build intelligent robots. A popular method to build intelligent robots is X. However, X has problem Y."

2. It should then contain a specific formulation of your research question (RQ). For a good description of how to formulate a research question, consider `https://www.scribbr.com/research-process/research-questions/`. E.g.: "Problem Y has not yet been addressed sufficiently by other researchers, so how can we extend X to address Y?"

3. In addition to your research question, you can (but don't have to) state a specific hypothesis: "I hypothesize that we can address Y by extending X with Z." It is important that you also say how you plan to validate your hypothesis. For example, if you do an empirical investigation, write something like "I will validate my hypothesis by measuring ... with experiment ...". If you write a more formal thesis, you might want to prove a certain theorem, so you could write "I will validate my hypothesis by proving that ..."

4. The RQ and hypothesis should be followed by a very brief (one-two sentences) summary of how you want to address the question, i.e., the methods and

evaluation metrics you want to use, highlighting your research contributions with respect to the state of the art. Herein, you should also refer to the individual sections in your paper to give the reader a brief outline. "The review of the related work in Section 2 of this paper shows that method Q has not yet been investigated as a solution to problem X. As a novel contribution, I will address this hypothesis / RQ by implementing Z using method Q, as described in Section 3. I will evaluate the results by measuring V, as described in Section 4." (Note that you might not have a Section 4 if you do not conduct empirical research but focus on theory instead.)

5. You should conclude with a sentence like this: "I conclude in Section 5, by highlighting that extending X with Z indeed achieves Y, but only if W holds."

It does not matter whether you use "I" or "We". It does matter, however, that you use the same pronoun consistently.

Chapter 2

Preliminaries

2.1 Control Theory Basics

Information about state space models, LQR, etc.

2.2 Notation

Which letters I will use in what context

Chapter 3

Background

3.1 Motivation

Reviewing content of past papers, state of the art, and what the open problems are.

3.2 Approach

What I will do in the following sections

Chapter 4

Robustness of LQR with low-dimensional Input

4.1 Setting

4.2 Results

4.3 Discussion

Chapter 5

Robustness under artificially inflated Input

5.1 Setting

5.2 Results

5.3 Discussion

Chapter 6

Perception-based Experiments

6.1 Setting

6.2 Results

6.3 Discussion

Chapter 7

Conclusion

You generally cover three things in the Conclusions section, and each of these usually merits a separate subsection:

1. Conclusions
2. Summary of Contributions
3. Future Research

Conclusions are not a rambling summary of the thesis: they are short, concise statements of the inferences that you have made because of your work. It helps to organize these as short numbered paragraphs, ordered from most to least important. All conclusions should be directly related to the research question stated in the introduction. Examples:

1. The problem stated in the introduction has been solved: as shown in Sections ? to ??, an algorithm capable of handling large-scale Zylon problems in reasonable time has been developed.
2. The principal mechanism needed in the improved Zylon algorithm is the Grooty mechanism.
3. Etc.

The Summary of Contributions will be much sought and carefully read by the examiners. Here you list the contributions of new knowledge that your thesis makes. Of course, the thesis itself must substantiate any claims made here. There is often some overlap with the Conclusions, but that's okay. Concise numbered paragraphs are again best. Organize from most to least important. Examples:

1. Developed a much quicker algorithm for large-scale Zylon problems.
2. Demonstrated the first use of the Grooty mechanism for Zylon calculations.
3. Etc.

The Future Research subsection is included so that researchers picking up this work in future have the benefit of the ideas that you generated while you were working on the project. Again, concise numbered paragraphs are usually best.

Appendix A

Nomenclature

What goes in the appendices? Any material which impedes the smooth development of your presentation, but which is important to justify the results of a thesis. Generally it is material that is of too nitty-gritty a level of detail for inclusion in the main body of the thesis, but which should be available for perusal by the examiners to convince them sufficiently. Examples include program listings, immense tables of data, lengthy mathematical proofs or derivations, etc.

Appendix B

Additional Proofs

Appendix C

Complete Simulation Results

Bibliography

Erklärung der Urheberschaft

Ich versichere an Eides statt, dass ich die Bachelor's Thesis im Studiengang Data Science selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel – insbesondere keine im Quellenverzeichnis nicht benannten Internet-Quellen – benutzt habe. Es wurden keine Quellen genutzt, die nicht genannt wurden. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen entnommen wurden, sind als solche kenntlich gemacht. Ich versichere weiterhin, dass ich die Arbeit vorher nicht in einem anderen Prüfungsverfahren eingereicht habe und die eingereichte schriftliche Fassung der auf dem elektronischen Speichermedium entspricht.

Ort, Datum Unterschrift

Erklärung zur Veröffentlichung

Ich erkläre mein Einverständnis mit der Einstellung dieser Bachelor's Thesis in den Bestand der Bibliothek.

Ort, Datum Unterschrift